AUDIT REPORT

hcp-graph-targeting: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/hcp-graph-targeting
Numbers taken from commit	`719ffbfc4cf1d5bf57f19f9c80be25168ad972f6` (outputs last changed in `719ffbf (2026-10-13 21:46:05 +0400)`)
Video	`docs/demo/demo.mp4` · 88.0 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:15
Automated checks	57 pass · 0 fail · 2 manual (of 59 figures)
Assumptions	12 (A-01 to A-12)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                  # data/ + outputs/ablation.csv, deltas.csv,
feature_importance.csv, summary.json
uv run python scripts/seed_sweep.py       # outputs/seed_sweep.csv/.txt (seeds 11, 23)
.venv/bin/python docs/demo/source/tools/export_call_plan.py docs/demo/source/assets/call_plan.json   # ~30 s;
asserts AUC + captured match ablation.csv
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/call_plan.json
docs/demo/source/compositions --rows 1,2,3,4,5,6
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q
looks -o renders/hcp-graph-targeting-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/hcp-graph-targeting-master.mp4 -c:v libx264 -preset slow -crf 25
-pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks,
MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 11-26	-	general knowledge	✓ verified
F-03	20,000 doctors / 20,000 HCPs	02, 05, 07, 10	outputs/summary.json	n_hcps	-	direct	✓ verified
F-04	75,155 ties	02, 05	outputs/summary.json; data/edges.csv	n_edges	-	direct	✓ verified
F-05	Calls for just 1 in 20	02	outputs/summary.json	budgets.captured_5pct = 1,000; n_hcps = 20,000	20,000 / 1,000 = 20	derived	✓ verified
F-06	8 specialties · 12 regions	02	data/hcps.csv	specialty, region (distinct values)	-	direct	✓ verified
F-07	24 months of class prescriptions	02, 05	data/monthly_class_rx.csv	month columns (2024-07..2026-06)	columns - hcp_id	direct	✓ verified
F-08	only 8.1% convert to the new drug / all 20,000 HCPs: 8.1%	02, 06	outputs/summary.json	conversion_rate = 0.08085 (1,617 of 20,000)	rounded to 1 decimal	direct	✓ verified
F-09	PRACTICES · groups of about 50	03	outputs/summary.json; README line 161	communities_per_level[1] = 408	20,000 / 408 = 49	derived	✓ verified
F-10	EVERY FOLD, 5 TIMES / Stratified 5-fold	03, 05	outputs/summary.json	folds	-	direct	✓ verified
F-11	more converters · shuffle-tested · leak-tested	03	outputs/ablation.csv; tests/test_leakage.py	captured_5pct graph features > decile list; control rows present; leakage tests exist	189 > 126	derived	✓ verified
F-12	held-out AUC fell to 0.589	04	docs/adr/0001-graph-features-built-fold-safe-embeddings-label-free.md:17	ADR text	-	direct	✓ verified
F-13	0.668	04, 08	outputs/ablation.csv	arm='+ community and neighbour conversion rates', auc	-	direct	✓ verified
F-14	and a test pins it	04	tests/test_leakage.py	own-label flip test	-	direct	✓ verified
F-15	PRACTICE RATE 2 / 5	04	schematic	-	-	illustrative	manual: Illustrative drawing of features.py label_graph_features; not real HCPs.
F-16	generate(), profile_features(), louvain_communities(), structural_features(), label_graph_features(), node2vec, GraphSAGE, make_folds(), run_arm(), captured_at_budget(), summarise(), bootstrap_delta(), cli.py shuffled control	05	src/hcp_targeting/	def names	-	direct	✓ verified
F-17	35 regions · 408 communities	05	outputs/summary.json	communities_per_level	-	direct	✓ verified
F-18	betweenness k=256	05	src/hcp_targeting/cli.py:198	--betweenness-k default	-	direct	✓ verified
F-19	400 trees · depth 4	05	src/hcp_targeting/evaluate.py:88-90	n_estimators, max_depth	-	direct	✓ verified
F-20	nine model arms · 7 arms + 2 controls · 45 fits	05	src/hcp_targeting/evaluate.py (ARMS); cli.py:98 (ARMS[2], ARMS[5]); outputs/ablation.csv	7 Arm(...) + 2 control rows	(7 + 2) x 5 folds = 45	derived	✓ verified
F-21	32-d walks · 2 × SAGEConv	05	src/hcp_targeting/cli.py:197; embeddings.py:131-133	--dim 32; conv1, conv2 = SAGEConv	-	direct	✓ verified
F-22	k = 1,000 · 4,000	05	outputs/summary.json	budgets	-	direct	✓ verified
F-23	200 resamples · 95% CI	05, 08	src/hcp_targeting/cli.py:201; evaluate.py:52	--n-boot default 200; percentiles 2.5/97.5	-	direct	✓ verified
F-24	full run ≈ 403 s · 10-core laptop	05	README line 178	README text	-	direct	✓ verified
F-25	CPU only, no GPU or API key / RUNS ON A LAPTOP CPU	05, 10	README 'Running it'	README text	-	direct	✓ verified
F-26	fallback: decile, then annual Rx (today's list)	05	src/hcp_targeting/cli.py:83-84	rule = decile * 1e6 + annual_class_rx	-	direct	✓ verified
F-27	practice + peers · no own label · train labels only	05	src/hcp_targeting/features.py:153-184; tests/test_leakage.py	inner_folds = 5	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-28	generate() · seed 7	05	outputs/summary.json	seed	-	direct	✓ verified
F-29	590 of 1,834 HCPs in one regional block · 1,304 ties	06	demo assets/call_plan.json (network_sample), exported by tools/export_call_plan.py from Louvain on data/edges.csv	len(nodes), l0_size, len(edges)	8 largest level-1 communities of the 5th-largest level-0 block	direct	✓ verified
F-30	9 of 59 15.3% · 7 of 68 10.3% · 6 of 60 10.0% · 4 of 64 6.2% · 4 of 87 4.6% · 4 of 97 4.1% · 2 of 78 2.6% · 2 of 79 2.5%	06	demo assets/call_plan.json (network_sample.communities); labels from data/hcps.csv converted	converted, size, rate	rate = converted / size	direct	✓ verified
F-31	● converted · ● not yet	06	data/hcps.csv; demo assets/call_plan.json	converted per hcp_id	-	direct	✓ verified
F-32	126 · 0.623	06	outputs/ablation.csv	arm='decile list (no model)': captured_5pct, auc	-	direct	✓ verified
F-33	166 · 0.641	06	outputs/ablation.csv	captured_5pct, auc	-	direct	✓ verified
F-34	188 · 0.658	06	outputs/ablation.csv	captured_5pct, auc	-	direct	✓ verified
F-35	189 · 0.668	06	outputs/ablation.csv	captured_5pct, auc	-	direct	✓ verified
F-36	202 · 0.676	06	outputs/ablation.csv	captured_5pct, auc	-	direct	✓ verified
F-37	212 · 0.674	06, 08	outputs/ablation.csv	captured_5pct, auc	-	direct	✓ verified
F-38	146 · 0.633	06, 08	outputs/ablation.csv	arm='control, HCPs shuffled across the graph: + community and neighbour conversion rates'	-	direct	✓ verified
F-39	322 · 0.761	06, 08	outputs/ablation.csv	arm='ceiling: true propensity (answer key)'	-	direct	✓ verified
F-40	#1 HCP 16728 … #6 HCP 04338	07	demo assets/call_plan.json, injected into compositions/value.html	plan ranks 1-6	score = out-of-fold p(convert); practice rate = fold-built community_conv_rate_l1 as the HCP saw it when held out; PEERS ✓ = nbr_converted	direct	✓ verified
F-41	+ 994 more on this plan	07	demo assets/call_plan.json	calls - 6 rows	1,000 - 6 = 994	derived	✓ verified
F-42	848 of the 1,000 are not in the decile list's top 1,000	07	demo assets/call_plan.json	plan[].decile_rank > 1,000	count	derived	✓ verified
F-43	1,000 calls · 189 converters reached · decile list: 126	07	demo assets/call_plan.json; outputs/ablation.csv	captured, captured_decile_list (equal to ablation.csv captured_5pct)	-	direct	✓ verified
F-44	29 of 100 HCPs here are planned · called: 29 · switched: 6	07	demo assets/call_plan.json (scene)	size, planned, planned_converted	-	direct	✓ verified
F-45	TOP 1,000 OF 20,000 HCPS	07	demo assets/call_plan.json	calls, n_hcps	-	direct	✓ verified
F-46	+50% · decile list 126 → 189	08	outputs/ablation.csv	captured_5pct: graph features 189, decile list 126	189 / 126 - 1 = +50.0%	derived	✓ verified
F-47	0.641 → 0.668 · +0.027 [+0.018, +0.037]	08	outputs/deltas.csv	arm='+ community and neighbour conversion rates', vs='profile + decile', metric=auc	-	direct	✓ verified
F-48	0.633 · below profile-only 0.641	08	outputs/ablation.csv	both control rows auc; profile + decile auc	-	direct	✓ verified
F-49	44% · 126 → 212 of 322	08	outputs/ablation.csv	captured_5pct: decile 126, best (n2v + SAGE) 212, ceiling 322	(212 - 126) / (322 - 126) = 43.9%	derived	✓ verified
F-50	+0.025 to +0.039 AUC	08	outputs/seed_sweep.csv	auc: 'graph features' minus 'profile' per seed (7, 11, 23)	min/max of 0.0269, 0.0247, 0.0388	derived	✓ verified
F-51	+0.027	09	outputs/deltas.csv	arm='+ community and neighbour conversion rates', vs='profile + decile', auc	-	direct	✓ verified
F-52	+0.008	09	outputs/deltas.csv	arm='graph features + node2vec', vs='graph features', auc	-	direct	✓ verified
F-53	-0.002	09	outputs/deltas.csv	arm='graph features + GraphSAGE', vs='graph features', auc	-	direct	✓ verified
F-54	+0.005	09	outputs/deltas.csv	arm='graph features + node2vec + GraphSAGE', vs='graph features', auc	-	direct	✓ verified
F-55	-0.010	09	outputs/deltas.csv	arm='profile + embeddings only (no hand-built graph features)', vs='graph	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
				features', auc			
F-56	61% · node2vec 40% · SAGE 21%	09	outputs/summary.json	gain_share_by_block_full_model: node2vec 0.3975, graphsage 0.2091	0.3975 + 0.2091 = 0.6067	derived	✓ verified
F-57	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-58	XGBoost · NetworkX · PyTorch Geometric · Python	10	pyproject.toml dependencies	xgboost, networkx, torch-geometric	-	direct	✓ verified
F-59	$ make run # 20,000 HCPs, 9 arms, 5 folds	10	Makefile; cli.py defaults	run target; --n-hcps 20000; 7 arms + 2 controls; --folds 5	-	direct	✓ verified

2. Automated check run (2026-10-14 14:15)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	20,000	20,000	✓ pass
F-04	75,155	75,155	✓ pass
F-05	1 in 20	1 in 20	✓ pass
F-06	true	true	✓ pass
F-07	24	24	✓ pass
F-08	8.1%	8.1%	✓ pass
F-09	49	49	✓ pass
F-10	5	5	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	0.668	0.668	✓ pass
F-14	true	true	✓ pass
F-15	PRACTICE RATE 2 / 5	-	manual
F-16	true	true	✓ pass
F-17	35 · 408	35 · 408	✓ pass
F-18	true	true	✓ pass
F-19	true	true	✓ pass
F-20	45	45	✓ pass
F-21	true	true	✓ pass
F-22	1,000 · 4,000	1,000 · 4,000	✓ pass
F-23	true	true	✓ pass
F-24	true	true	✓ pass
F-25	true	true	✓ pass
F-26	true	true	✓ pass
F-27	true	true	✓ pass
F-28	7	7	✓ pass
F-29	590 of 1,834 · 1,304	590 of 1,834 · 1,304	✓ pass
F-30	9 of 59 15.3% · 7 of 68 10.3% · 6 of 60 10.0% · 4 of 64 6.2% · 4 of 87 4.6% · 4 of 97 4.1% · 2 of 78 2.6% · 2 of 79 2.5%	9 of 59 15.3% · 7 of 68 10.3% · 6 of 60 10.0% · 4 of 64 6.2% · 4 of 87 4.6% · 4 of 97 4.1% · 2 of 78 2.6% · 2 of 79 2.5%	✓ pass
F-31	true	true	✓ pass
F-32	126 · 0.623	126 · 0.623	✓ pass
F-33	166 · 0.641	166 · 0.641	✓ pass
F-34	188 · 0.658	188 · 0.658	✓ pass
F-35	189 · 0.668	189 · 0.668	✓ pass
F-36	202 · 0.676	202 · 0.676	✓ pass
F-37	212 · 0.674	212 · 0.674	✓ pass
F-38	146 · 0.633	146 · 0.633	✓ pass
F-39	322 · 0.761	322 · 0.761	✓ pass
F-40	true	true	✓ pass
F-41	994	994	✓ pass
F-42	848	848	✓ pass
F-43	189 · 126	189 · 126	✓ pass
F-44	29 of 100 · 6	29 of 100 · 6	✓ pass
F-45	1,000 of 20,000	1,000 of 20,000	✓ pass
F-46	+50%	+50%	✓ pass
F-47	+0.027 [+0.018, +0.037]	+0.027 [+0.018, +0.037]	✓ pass
F-48	0.633 · 0.633 · 0.641	0.633 · 0.633 · 0.641	✓ pass
F-49	44%	44%	✓ pass
F-50	+0.025 to +0.039	+0.025 to +0.039	✓ pass
F-51	+0.027 [+0.018, +0.037]	+0.027 [+0.018, +0.037]	✓ pass
F-52	+0.008 [-0.000, +0.014]	+0.008 [-0.000, +0.014]	✓ pass
F-53	-0.002 [-0.008, +0.003]	-0.002 [-0.008, +0.003]	✓ pass
F-54	+0.005 [-0.002, +0.012]	+0.005 [-0.002, +0.012]	✓ pass
F-55	-0.010 [-0.020, -0.001]	-0.010 [-0.020, -0.001]	✓ pass
F-56	61% · 40% · 21%	61% · 40% · 21%	✓ pass
F-57	true	true	✓ pass

ID	Expected on slide	Recomputed from repo	Result
F-58	true	true	✓ pass
F-59	true	true	✓ pass

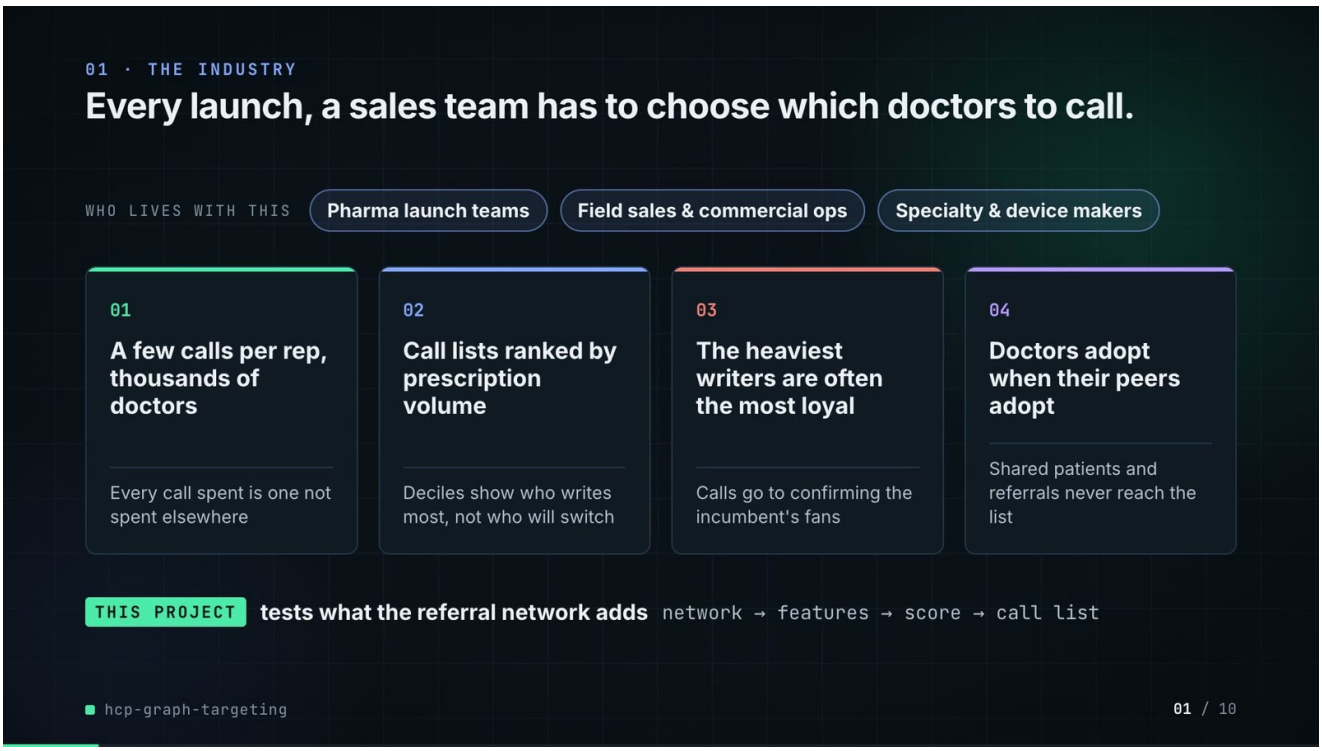
3. Assumptions register

ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator (20,000 HCPs, 8 specialties, 12 regions, about 440 practice groups, 24 months of class prescriptions, 75,155 ties). Real HCP prescribing and claims data is proprietary and not used.	src/hcp_targeting/data.py:78-206 (generate, seed 7); scripts/generate_data.py; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	The generator decides how much the network matters: conversion depends on hidden practice-group culture (sd 0.75 log-odds) and exposure to advocate key opinion leaders (+1.3 log-odds at full exposure). The size of the graph lift is a property of the generator, not evidence about a real market.	src/hcp_targeting/data.py:96-98, :181-191; README 'Known limitations' (biggest gap)	F-13, F-29, F-30, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-43, F-46, F-47, F-48, F-49, F-50	slides 04, 06, 07, 08
A-03	A call budget is 5% (1,000 calls) or 20% (4,000 calls) of the 20,000 HCPs; 'converters reached' assumes the call is what converts the HCP (the model ranks propensity, not uplift from calling).	src/hcp_targeting/cli.py:199-200 (--budget-small 0.05, --budget-large 0.20); evaluate.py:23-26 (captured_at_budget); README 'Known limitations'	F-05, F-22, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-41, F-42, F-43, F-44, F-45, F-46, F-49	slides 02, 05, 06, 07, 08
A-04	Label-derived features (community and neighbour conversion rates) are rebuilt inside every outer fold from training labels only, with an inner 5-fold split and a prior weight of 20; embeddings are trained without labels.	src/hcp_targeting/features.py:153-184 (label_graph_features); evaluate.py:126-130; docs/adr/0001	F-10, F-11, F-12, F-13, F-27, F-35, F-40	slides 03, 04, 05, 06, 07, 08
A-05	Louvain runs at resolution 1 (regional blocks) and 16 (practice-size communities); 16 was chosen to match practice size, not tuned on conversion.	src/hcp_targeting/features.py:60-76; README line 161	F-09, F-17, F-29, F-30, F-44	slides 03, 05, 06, 07
A-06	Betweenness is estimated from 256 sampled sources (64 in the seed sweep).	src/hcp_targeting/cli.py:198; features.py:89-91; scripts/seed_sweep.py:47	F-18, F-50	slides 05, 08
A-07	Every arm uses one XGBoost setting (400 trees, depth 4, learning rate 0.03, seed 7) on the same 5 stratified folds; 7 arms plus 2 shuffled-graph controls = 45 fits.	src/hcp_targeting/evaluate.py:65-99, :143-144; cli.py:82-104	F-13, F-19, F-20, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-46, F-47, F-48	slides 04, 05, 06, 08
A-08	The shuffled-graph control moves every HCP to a random node of the same graph (seed 8 permutation): structure is unchanged, the person-position link is broken.	src/hcp_targeting/cli.py:93-104	F-11, F-38, F-48	slides 03, 06, 08
A-09	The ceiling row scores HCPs by the generator's true propensity (answer key), which no feature reads.	src/hcp_targeting/cli.py:106-107; data.py:42-43, :221-226	F-39, F-49	slides 06, 08
A-10	The call plan on slide 07 ranks the arm '+ community and neighbour conversion rates' (hand-built graph features, no embeddings) by its out-of-fold scores; the demo's export re-runs that arm and asserts AUC 0.668 and 189 / 126 converters captured match outputs/ablation.csv.	docs/demo/source/tools/export_call_plan.py	F-40, F-41, F-42, F-43, F-44, F-45	slides 07
A-11	Video rounding: AUC to 3 decimals, AUC deltas to 3 decimals with sign, percentages to 1 decimal (community rates, 8.1%) or whole numbers (+50%, 44%, 61%, 40%, 21%, practice rate on slide 07), counts with thousands separators; '1 in 20' = 5% budget; '≈403 s' quotes the README.	this audit (display convention)	F-05, F-08, F-30, F-46, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-56	slides 02, 06, 08, 09
A-12	Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, consistent with README 'The problem'. It carries no numbers and is not a repo result.	author's framing; README 'The problem'	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every launch, a sales team has to choose which doctors to call.

WHO LIVES WITH THIS · Pharma launch teams · Field sales & commercial ops · Specialty & device makers

01 A few calls per rep, thousands of doctors — Every call spent is one not spent elsewhere

02 Call lists ranked by prescription volume — Deciles show who writes most, not who will switch

03 The heaviest writers are often the most loyal — Calls go to confirming the incumbent's fans

04 Doctors adopt when their peers adopt — Shared patients and referrals never reach the list

THIS PROJECT tests what the referral network adds · network → features → score → call list

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 11-26	-	✓

Assumptions: A-12

Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

20,000 doctors. 75,155 ties between them. Calls for just 1 in 20.

8 specialties · 12 regions · 24 months of class prescriptions · only 8.1% convert to the new drug

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 doctors / 20,000 HCPs	`outputs/summary.json` · n_hcps	-	✓
F-04	75,155 ties	`outputs/summary.json; data/edges.csv` · n_edges	-	✓
F-05	Calls for just 1 in 20	`outputs/summary.json` · budgets.captured_5pct = 1,000; n_hcps = 20,000	20,000 / 1,000 = 20	✓
F-06	8 specialties · 12 regions	`data/hcps.csv` · specialty, region (distinct values)	-	✓
F-07	24 months of class prescriptions	`data/monthly_class_rx.csv` · month columns (2024-07..2026-06)	columns - hcp_id	✓
F-08	only 8.1% convert to the new drug / all 20,000 HCPs: 8.1%	`outputs/summary.json` · conversion_rate = 0.08085 (1,617 of 20,000)	rounded to 1 decimal	✓

Assumptions: A-01, A-03

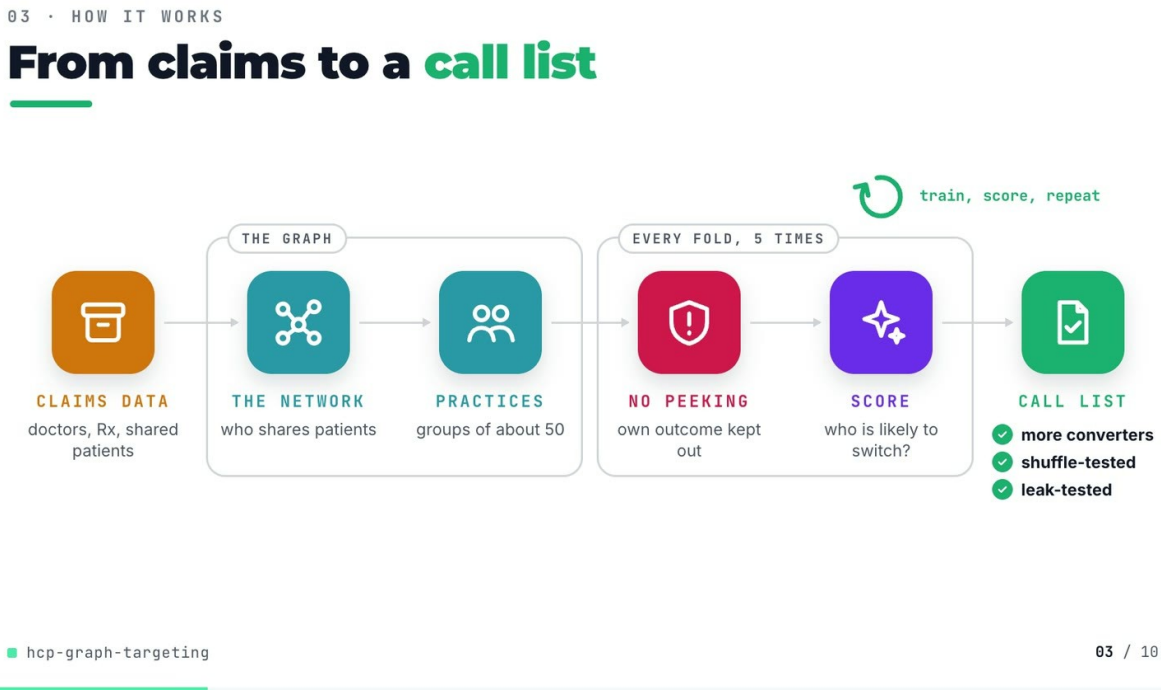
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html.

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From claims to a call list

CLAIMS DATA doctors, Rx, shared patients → [THE GRAPH: THE NETWORK who shares patients → PRACTICES groups of about 50] → [EVERY FOLD, 5 TIMES: NO PEEKING own outcome kept out → SCORE who is likely to switch?] → CALL LIST ✓ more converters ✓ shuffle-tested ✓ leak-tested

train, score, repeat (loop over the scoring step)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-09	PRACTICES · groups of about 50	`outputs/summary.json; README line 161` · communities_per_level[1] = 408	20,000 / 408 = 49	✓
F-10	EVERY FOLD, 5 TIMES / Stratified 5-fold	`outputs/summary.json` · folds	-	✓
F-11	more converters · shuffle-tested · leak-tested	`outputs/ablation.csv; tests/test_leakage.py` · captured_5pct graph features > decile list; control rows present; leakage tests exist	189 > 126	✓

Assumptions: A-04, A-05, A-08

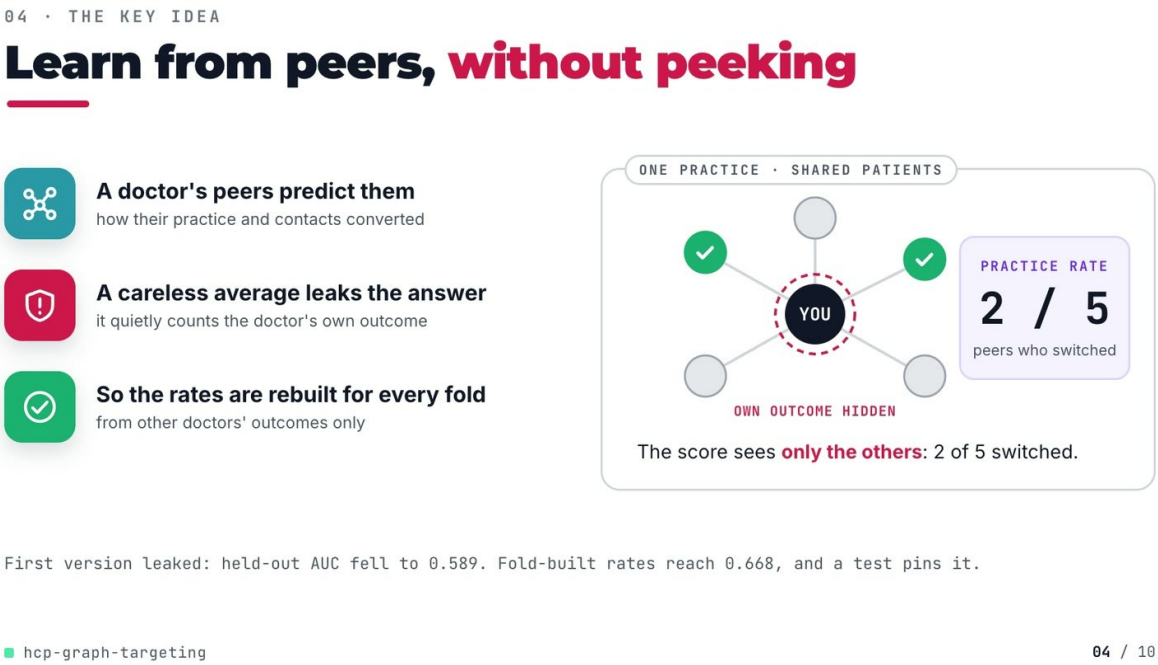
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'Claims data' describes where shared-patient ties come from in practice; the repo's data is generated (A-01).

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth is F-11.

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

Learn from peers, without peeking

A doctor's peers predict them — how their practice and contacts converted

A careless average leaks the answer — it quietly counts the doctor's own outcome

So the rates are rebuilt for every fold — from other doctors' outcomes only

ONE PRACTICE · SHARED PATIENTS: YOU + 5 peers, OWN OUTCOME HIDDEN, PRACTICE RATE 2 / 5 peers who switched. The score sees only the others: 2 of 5 switched.

First version leaked: held-out AUC fell to 0.589. Fold-built rates reach 0.668, and a test pins it.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-12	held-out AUC fell to 0.589	`docs/adr/0001-graph-features-built-fold-safe-embeddings-label-free.md:17` · ADR text	-	✓
F-13	0.668	`outputs/ablation.csv` · arm='+ community and neighbour conversion rates', auc	-	✓
F-14	and a test pins it	`tests/test_leakage.py` · own-label flip test	-	✓
F-15	PRACTICE RATE 2 / 5	`schematic` · -	-	manual

Assumptions: A-04, A-07

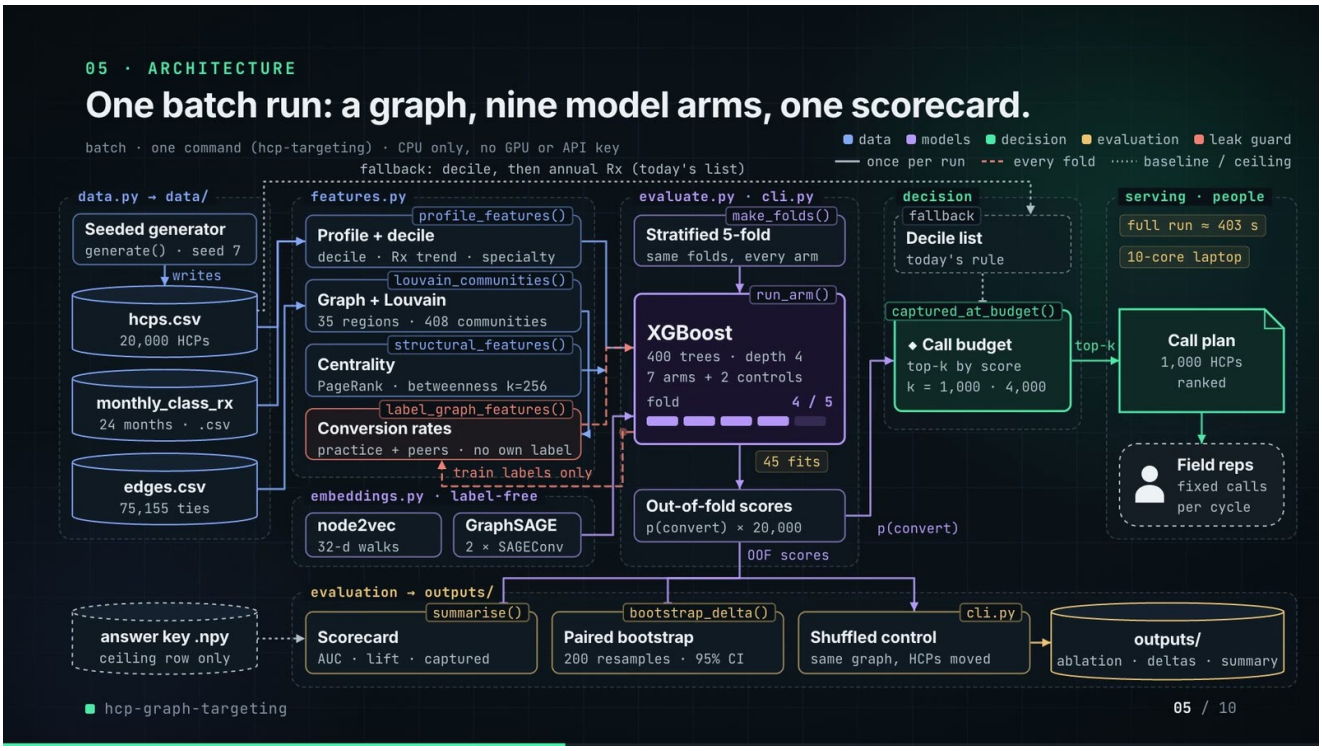
Illustrative, not data:

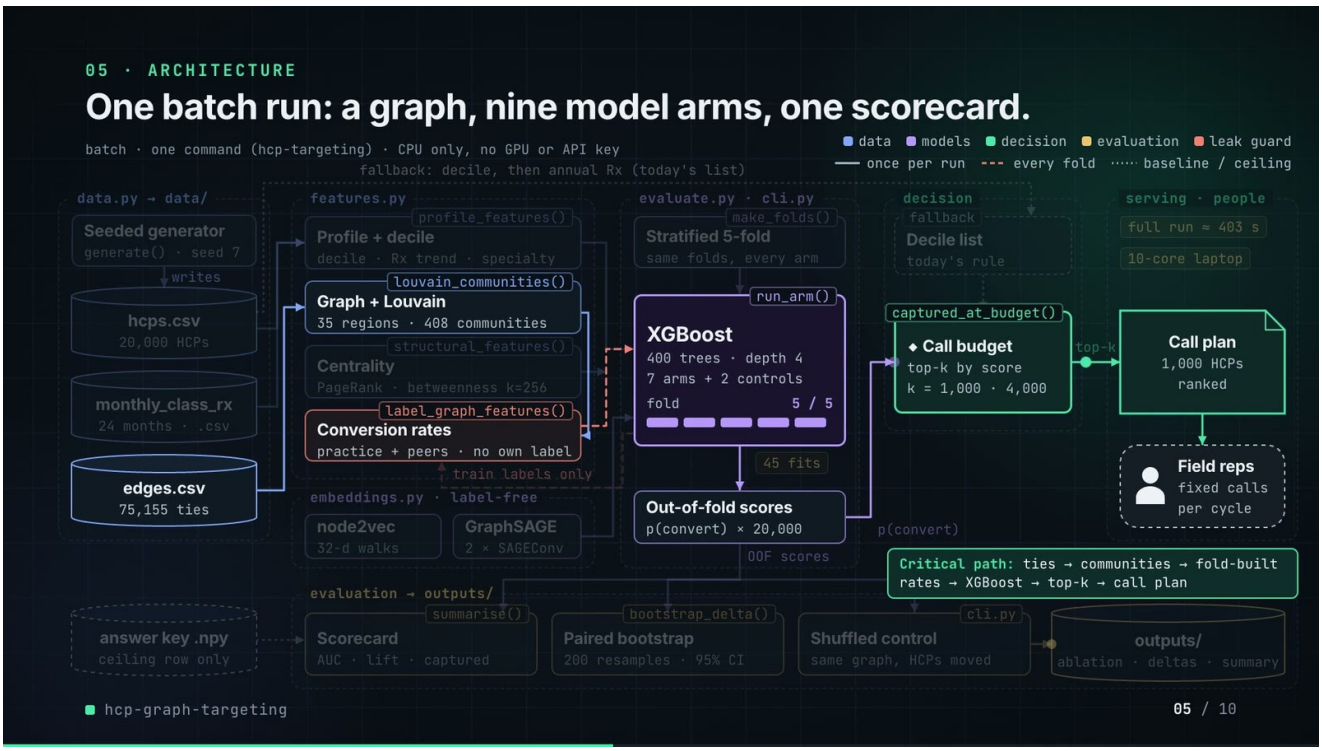
- The YOU + 5 peers demo is schematic (F-15); it mirrors label_graph_features, not real HCPs.

To change this slide: Markup under #ex-b and the practice demo block in compositions/explainer.html; the footnote figures are F-12, F-13, F-14.

Slide 05: Architecture

26.0-42.0 s · composition `compositions/s03-architecture.html`





On-screen text (verbatim)

05 · ARCHITECTURE

One batch run: a graph, nine model arms, one scorecard.

batch · one command (hcp-targeting) · CPU only, no GPU or API key

Legend: data · models · decision · evaluation · leak guard; once per run · every fold · baseline / ceiling

data.py → data/: Seeded generator (generate() · seed 7) writes hcps.csv (20,000 HCPs), monthly_class_rx (24 months · .csv), edges.csv (75,155 ties); answer key .npy (ceiling row only)

features.py: profile_features() Profile + decile (decile · Rx trend · specialty); louvain_communities() Graph + Louvain (35 regions · 408 communities); structural_features() Centrality (PageRank · betweenness k=256); label_graph_features() Conversion rates (practice + peers · no own label)

embeddings.py · label-free: node2vec (32-d walks), GraphSAGE (2 × SAGEConv)
evaluate.py · cli.py: make_folds() Stratified 5-fold (same folds, every arm) → run_arm() XGBoost (400 trees · depth 4 · 7 arms + 2 controls · fold n / 5) → Out-of-fold scores (p(convert) × 20,000); 45 fits; feedback loop 'train labels only' back to Conversion rates
decision: fallback Decile list (today's rule) → captured_at_budget() ◆ Call budget (top-k by score · k = 1,000 · 4,000)
serving · people: Call plan (1,000 HCPs ranked) → Field reps (fixed calls per cycle); notes: full run ≈ 403 s · 10-core laptop
evaluation → outputs/: summarise() Scorecard (AUC · lift · captured), bootstrap_delta() Paired bootstrap (200 resamples · 95% CI), cli.py Shuffled control (same graph, HCPs moved) → outputs/ (ablation · deltas · summary)
Edge labels: writes · train labels only · fallback: decile, then annual Rx (today's list) · p(convert) · top-k · OOF scores
Critical path: ties → communities → fold-built rates → XGBoost → top-k → call plan

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 doctors / 20,000 HCPs	`outputs/summary.json` · n_hcps	-	✓
F-04	75,155 ties	`outputs/summary.json; data/edges.csv` · n_edges	-	✓
F-07	24 months of class prescriptions	`data/monthly_class_rx.csv` · month columns (2024-07..2026-06)	columns - hcp_id	✓
F-10	EVERY FOLD, 5 TIMES / Stratified 5-fold	`outputs/summary.json` · folds	-	✓
F-16	generate(), profile_features(), louvain_communities(), structural_features(), label_graph_features(), node2vec, GraphSAGE, make_folds(), run_arm(), captured_at_budget(), summarise(), bootstrap_delta(), cli.py shuffled control	`src/hcp_targeting/` · def names	-	✓
F-17	35 regions · 408 communities	`outputs/summary.json` · communities_per_level	-	✓
F-18	betweenness k=256	`src/hcp_targeting/cli.py:198` · --betweenness-k default	-	✓
F-19	400 trees · depth 4	`src/hcp_targeting/evaluate.py:88-90` · n_estimators, max_depth	-	✓
F-20	nine model arms · 7 arms + 2 controls · 45 fits	`src/hcp_targeting/evaluate.py (ARMS); cli.py:98 (ARMS[2], ARMS[5]); outputs/ablation.csv` · 7 Arm(...) + 2 control rows	(7 + 2) x 5 folds = 45	✓
F-21	32-d walks · 2 × SAGEConv	`src/hcp_targeting/cli.py:197; embeddings.py:131-133` · --dim 32; conv1, conv2 = SAGEConv	-	✓
F-22	k = 1,000 · 4,000	`outputs/summary.json` · budgets	-	✓
F-23	200 resamples · 95% CI	`src/hcp_targeting/cli.py:201; evaluate.py:52` · --n-boot default 200; percentiles 2.5/97.5	-	✓
F-24	full run ≈ 403 s · 10-core laptop	`README line 178` · README text	-	✓
F-25	CPU only, no GPU or API key / RUNS ON A LAPTOP CPU	`README 'Running it'` · README text	-	✓
F-26	fallback: decile, then annual Rx (today's list)	`src/hcp_targeting/cli.py:83-84` · rule = decile * 1e6 + annual_class_rx	-	✓
F-27	practice + peers · no own label · train labels only	`src/hcp_targeting/features.py:153-184; tests/test_leakage.py` · inner_folds = 5	-	✓
F-28	generate() · seed 7	`outputs/summary.json` · seed	-	✓

F-20 note: 7 arms: T().count(' Arm(') counts the seven entries of ARMS (the dataclass definition line is 'class Arm:').

F-24 note: README states this for an earlier session; the captured run in outputs/ took 12,279 s (3.4 h) on a shared machine (summary.json runtime_seconds).

Assumptions: A-01, A-03, A-04, A-05, A-06, A-07, A-08, A-09

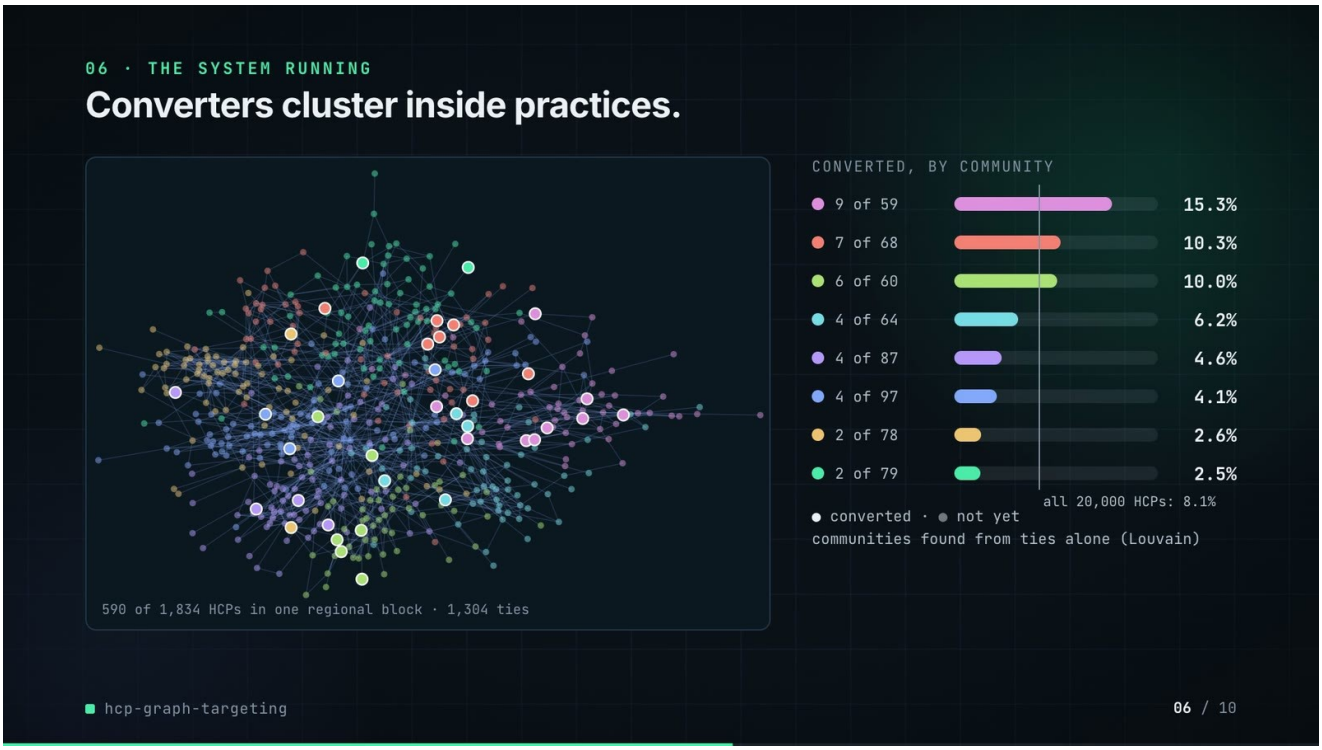
Illustrative, not data:

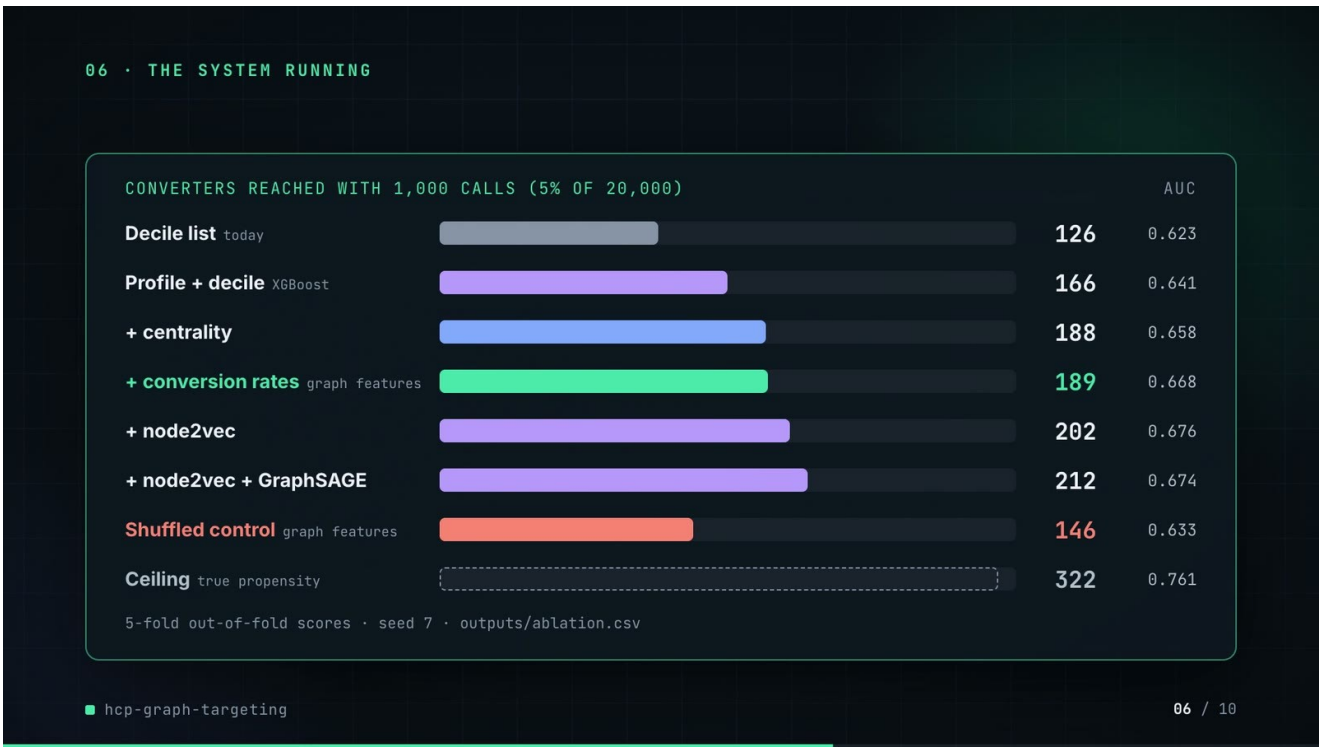
- Packets moving along the edges follow execution order but their speed is decorative; the fold counter shows the 5 folds, not wall-clock time.
- 'Field reps · fixed calls per cycle' is the human-in-the-loop consumer from the README problem statement; the repo has no rep or scheduling code.
- 'Call plan' is the top-k of the out-of-fold scores (what captured_at_budget counts); the repo does not write a plan file — the demo's export script does (slide 07).

To change this slide: FRAMES, N (components) and W (edges) objects in compositions/s03-architecture.html.

Slide 06: The system running

41.6-56.2 s · composition `compositions/s04-routes.html` (NET block injected by `tools/inject_sheet.py`; LADDER array)





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

Converters cluster inside practices.

Network panel: 590 of 1,834 HCPs in one regional block · 1,304 ties

CONVERTED, BY COMMUNITY: 9 of 59 15.3% · 7 of 68 10.3% · 6 of 60 10.0% · 4 of 64 6.2% · 4 of 87 4.6% · 4 of 97 4.1% · 2 of 78 2.6% · 2 of 79 2.5% · all 20,000 HCPs: 8.1%

● converted · ● not yet · communities found from ties alone (Louvain)

CONVERTERS REACHED WITH 1,000 CALLS (5% OF 20,000) · AUC: Decile list today 126 0.623 · Profile + decile XGBoost 166 0.641 · + centrality 188 0.658 · + conversion rates (graph features) 189 0.668 · + node2vec 202 0.676 · + node2vec + GraphSAGE 212 0.674 · Shuffled control (graph features) 146 0.633 · Ceiling (true propensity) 322 0.761

5-fold out-of-fold scores · seed 7 · outputs/ablation.csv

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-08	only 8.1% convert to the new drug / all 20,000 HCPs: 8.1%	`outputs/summary.json` · conversion_rate = 0.08085 (1,617 of 20,000)	rounded to 1 decimal	✓
F-29	590 of 1,834 HCPs in one regional block · 1,304 ties	`demo assets/call_plan.json (network_sample), exported by tools/export_call_plan.py from Louvain on data/edges.csv` · len(nodes), l0_size, len(edges)	8 largest level-1 communities of the 5th-largest level-0 block	✓
F-30	9 of 59 15.3% · 7 of 68 10.3% · 6 of 60 10.0% · 4 of 64 6.2% · 4 of 87 4.6% · 4 of 97 4.1% · 2 of 78 2.6% · 2 of 79 2.5%	`demo assets/call_plan.json (network_sample.communities); labels from data/hcps.csv converted` · converted, size, rate	rate = converted / size	✓
F-31	● converted · ● not yet	`data/hcps.csv; demo assets/call_plan.json` · converted per hcp_id	-	✓
F-32	126 · 0.623	`outputs/ablation.csv` · arm='decile list (no model)': captured_5pct, auc	-	✓
F-33	166 · 0.641	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-34	188 · 0.658	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-35	189 · 0.668	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-36	202 · 0.676	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-37	212 · 0.674	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-38	146 · 0.633	`outputs/ablation.csv` · arm='control, HCPs shuffled across the graph: + community and neighbour conversion rates'	-	✓
F-39	322 · 0.761	`outputs/ablation.csv` · arm='ceiling: true propensity (answer key)'	-	✓

F-30 note: Counts are over the whole level-1 community. One community (6 of 60) has 2 members outside the sampled regional block, so the panel draws 58 of its 60 HCPs; all 6 converters are inside. Node labels are re-checked against data/hcps.csv by sample_labels_match() (F-31).

Assumptions: A-02, A-03, A-05, A-07, A-08, A-09

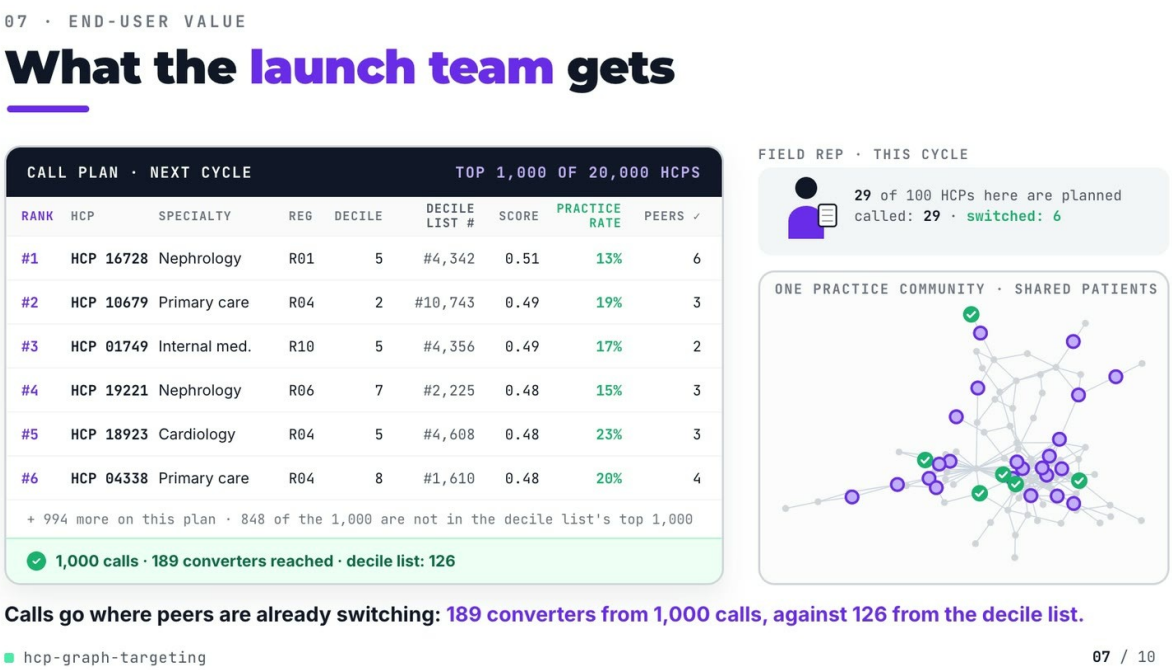
Illustrative, not data:

- The network layout is a seeded spring layout (networkx, seed 7) of the real sampled subgraph; position has no meaning beyond tie structure.
- Community colours are arbitrary.
- Bar lengths: 860 px track = 340 converters.

To change this slide: Re-run tools/export_call_plan.py + tools/inject_sheet.py for the network panel; LADDER array in compositions/s04-routes.html for the ladder.

Slide 07: End-user value (white)

55.9-68.9 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the launch team gets

CALL PLAN · NEXT CYCLE · TOP 1,000 OF 20,000 HCPS — columns RANK, HCP, SPECIALTY, REG, DECILE, DECILE LIST #, SCORE, PRACTICE RATE, PEERS ✓

#1 HCP 16728 Nephrology R01 5 #4,342 0.51 13% 6

#2 HCP 10679 Primary care R04 2 #10,743 0.49 19% 3

#3 HCP 01749 Internal med. R10 5 #4,356 0.49 17% 2

#4 HCP 19221 Nephrology R06 7 #2,225 0.48 15% 3

#5 HCP 18923 Cardiology R04 5 #4,608 0.48 23% 3

#6 HCP 04338 Primary care R04 8 #1,610 0.48 20% 4

+ 994 more on this plan · 848 of the 1,000 are not in the decile list's top 1,000

✓ 1,000 calls · 189 converters reached · decile list: 126

FIELD REP · THIS CYCLE: 29 of 100 HCPs here are planned · called: 29 · switched: 6

ONE PRACTICE COMMUNITY · SHARED PATIENTS

Calls go where peers are already switching: 189 converters from 1,000 calls, against 126 from the decile list.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 doctors / 20,000 HCPs	`outputs/summary.json` · n_hcps	-	✓
F-40	#1 HCP 16728 … #6 HCP 04338	`demo assets/call_plan.json, injected into compositions/value.html` · plan ranks 1-6	score = out-of-fold p(convert); practice rate = fold-built community_conv_rate_l1 as the HCP saw it when held out; PEERS ✓ = nbr_converted	✓
F-41	+ 994 more on this plan	`demo assets/call_plan.json` · calls - 6 rows	1,000 - 6 = 994	✓
F-42	848 of the 1,000 are not in the decile list's top 1,000	`demo assets/call_plan.json` · plan[].decile_rank > 1,000	count	✓
F-43	1,000 calls · 189 converters reached · decile list: 126	`demo assets/call_plan.json`; `outputs/ablation.csv` · captured, captured_decile_list (equal to ablation.csv captured_5pct)	-	✓
F-44	29 of 100 HCPs here are planned · called: 29 · switched: 6	`demo assets/call_plan.json` (scene) · size, planned, planned_converted	-	✓
F-45	TOP 1,000 OF 20,000 HCPS	`demo assets/call_plan.json` · calls, n_hcps	-	✓

Assumptions: A-02, A-03, A-04, A-10

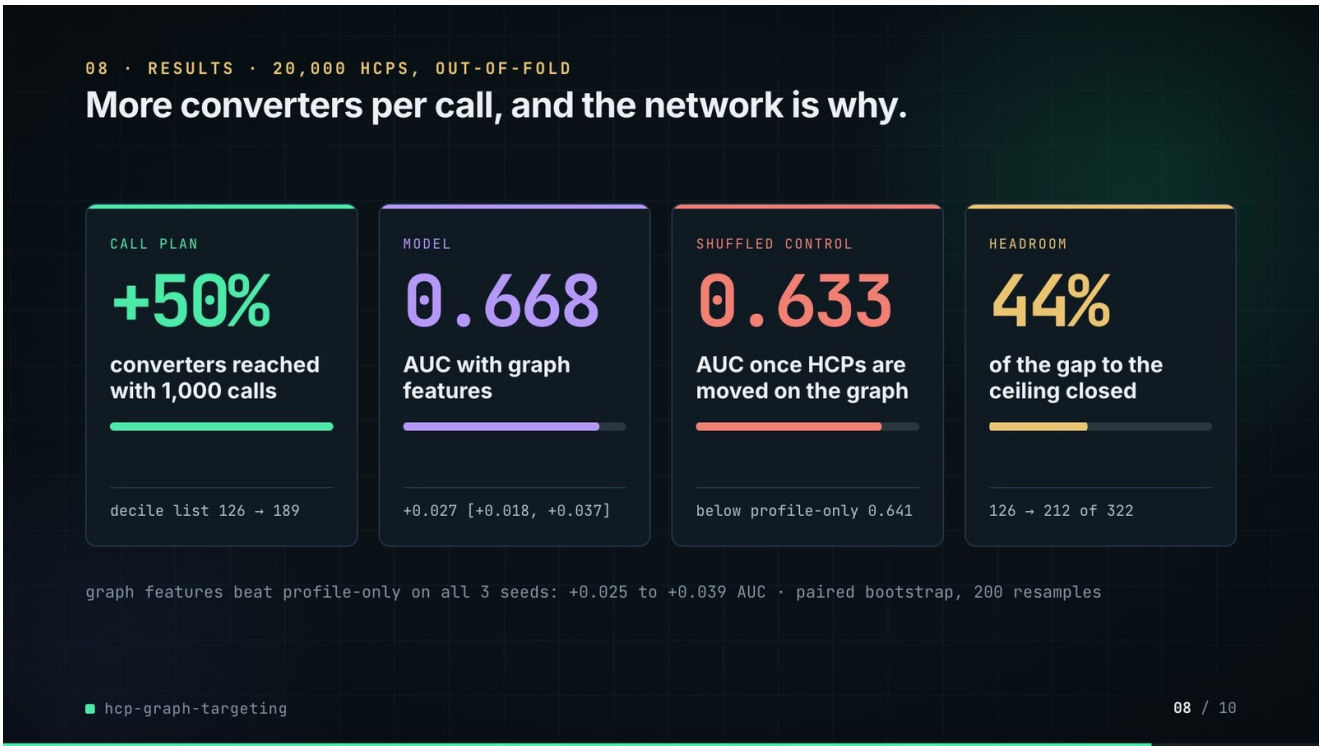
Illustrative, not data:

- HCP numbers are the generator's synthetic hcp_id values (no real people).
- 'NEXT CYCLE' frames the out-of-fold ranking as a plan; in the repo the scores and outcomes come from the same campaign window (README 'Known limitations': static snapshot).
- The rep figure, call cards and their timing are illustrative; the community (level-1 community 8), its ties, which members are on the plan and which converted are real, laid out with a seeded spring layout.
- Practice rate is the smoothed fold-built community conversion rate (prior weight 20), so it is not a raw share.

To change this slide: Re-run tools/export_call_plan.py, then tools/inject_sheet.py (--rows picks the plan ranks); layout in compositions/value.html.

Slide 08: Results

68.6-77.4 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · 20,000 HCPS, OUT-OF-FOLD

More converters per call, and the network is why.

CALL PLAN +50% converters reached with 1,000 calls · decile list 126 → 189

MODEL 0.668 AUC with graph features · +0.027 [+0.018, +0.037]

SHUFFLED CONTROL 0.633 AUC once HCPs are moved on the graph · below profile-only 0.641

HEADROOM 44% of the gap to the ceiling closed · 126 → 212 of 322

graph features beat profile-only on all 3 seeds: +0.025 to +0.039 AUC · paired bootstrap, 200 resamples

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	0.668	`outputs/ablation.csv` · arm='+ community and neighbour conversion rates', auc	-	✓
F-23	200 resamples · 95% CI	`src/hcp_targeting/cli.py:201; evaluate.py:52` · --n-boot default 200; percentiles 2.5/97.5	-	✓
F-37	212 · 0.674	`outputs/ablation.csv` · captured_5pct, auc	-	✓
F-38	146 · 0.633	`outputs/ablation.csv` · arm='control, HCPs shuffled across the graph: + community and neighbour conversion rates'	-	✓
F-39	322 · 0.761	`outputs/ablation.csv` · arm='ceiling: true propensity (answer key)'	-	✓
F-46	+50% · decile list 126 → 189	`outputs/ablation.csv` · captured_5pct: graph features 189, decile list 126	189 / 126 - 1 = +50.0%	✓
F-47	0.641 → 0.668 · +0.027 [+0.018, +0.037]	`outputs/deltas.csv` · arm='+ community and neighbour conversion rates', vs='profile + decile', metric=auc	-	✓
F-48	0.633 · below profile-only 0.641	`outputs/ablation.csv` · both control rows auc; profile + decile auc	-	✓
F-49	44% · 126 → 212 of 322	`outputs/ablation.csv` · captured_5pct: decile 126, best (n2v + SAGE) 212, ceiling 322	(212 - 126) / (322 - 126) = 43.9%	✓
F-50	+0.025 to +0.039 AUC	`outputs/seed_sweep.csv` · auc: 'graph features' minus 'profile' per seed (7, 11, 23)	min/max of 0.0269, 0.0247, 0.0388	✓

Assumptions: A-02, A-03, A-07, A-08, A-09, A-11

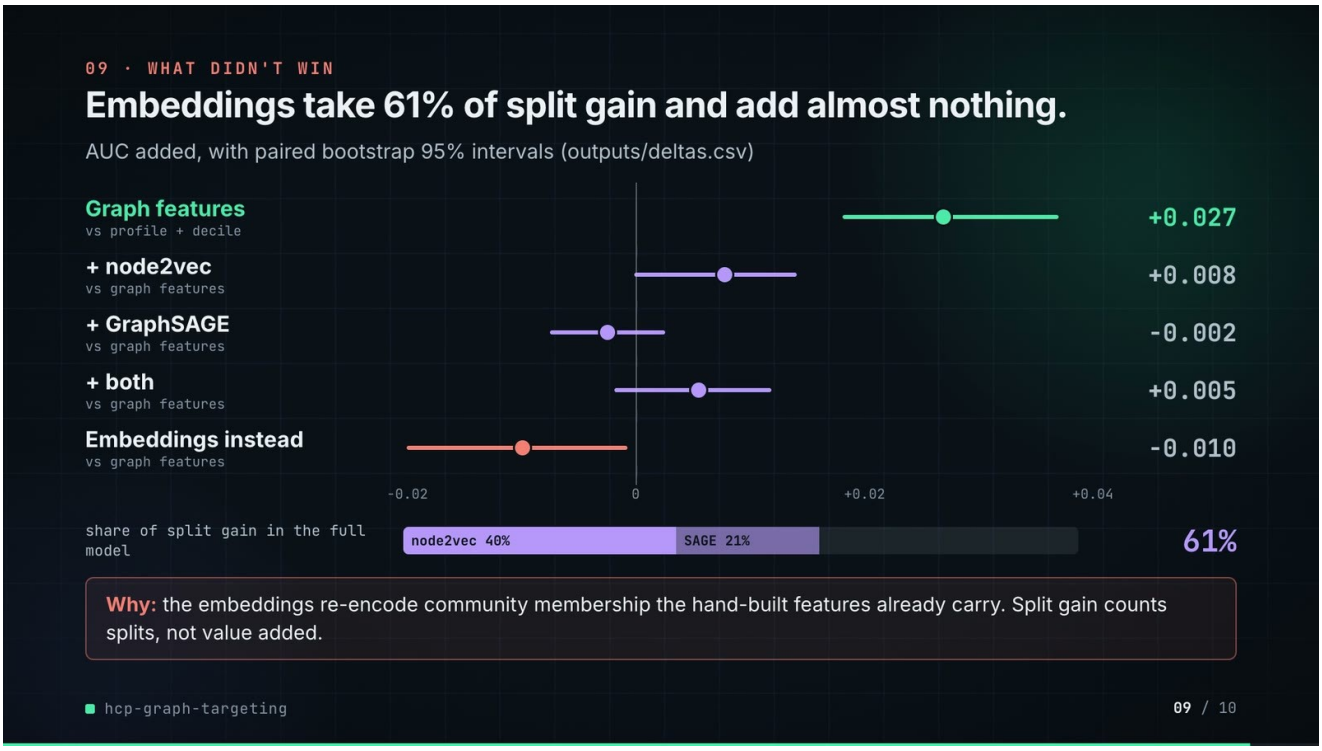
Illustrative, not data:

- Mini bars: card 1 and 4 are drawn to the ratio shown; cards 2-3 are AUC / 0.761 (ceiling).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

77.0-84.0 s · composition `compositions/s06-twist.html (ROWS array)`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

Embeddings take 61% of split gain and add almost nothing.

AUC added, with paired bootstrap 95% intervals (outputs/deltas.csv)

Graph features vs profile + decile +0.027 · + node2vec vs graph features +0.008 · + GraphSAGE vs graph features -0.002 · + both vs graph features +0.005 · Embeddings instead vs graph features -0.010

share of split gain in the full model: node2vec 40% · SAGE 21% · 61%

Why: the embeddings re-encode community membership the hand-built features already carry. Split gain counts splits, not value added.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-51	+0.027	`outputs/deltas.csv` · arm='+ community and neighbour conversion rates', vs='profile + decile', auc	-	✓
F-52	+0.008	`outputs/deltas.csv` · arm='graph features + node2vec', vs='graph features', auc	-	✓
F-53	-0.002	`outputs/deltas.csv` · arm='graph features + GraphSAGE', vs='graph features', auc	-	✓
F-54	+0.005	`outputs/deltas.csv` · arm='graph features + node2vec + GraphSAGE', vs='graph features', auc	-	✓
F-55	-0.010	`outputs/deltas.csv` · arm='profile + embeddings only (no hand-built graph features)', vs='graph features', auc	-	✓
F-56	61% · node2vec 40% · SAGE 21%	`outputs/summary.json` · gain_share_by_block_full_model: node2vec 0.3975, graphsage 0.2091	0.3975 + 0.2091 = 0.6067	✓

F-53 note: delta = -0.00249; the README rounds it to -0.003, the video shows -0.002 (3-decimal rounding of the CSV value).

F-54 note: delta = 0.00549; the README writes +0.006.

Assumptions: A-02, A-07, A-11

Illustrative, not data:

- Axis -0.02 … +0.04 AUC; dots are point estimates, lines are the 95% intervals.

To change this slide: ROWS array (deltas) and the gain bar in compositions/s06-twist.html.

Slide 10: End card

83.6-88.0 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP CPU

hcp-graph-targeting

github.com/kavya2693/hcp-graph-targeting

XGBoost · NetworkX · PyTorch Geometric · Python

$ make run # 20,000 HCPs, 9 arms, 5 folds

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 doctors / 20,000 HCPs	`outputs/summary.json` · n_hcps	-	✓
F-25	CPU only, no GPU or API key / RUNS ON A LAPTOP CPU	`README 'Running it'` · README text	-	✓
F-57	MIT	`LICENSE line 1` · -	-	✓
F-58	XGBoost · NetworkX · PyTorch Geometric · Python	`pyproject.toml dependencies` · xgboost, networkx, torch-geometric	-	✓
F-59	$ make run # 20,000 HCPs, 9 arms, 5 folds	`Makefile; cli.py defaults` · run target; --n-hcps 20000; 7 arms + 2 controls; --folds 5	-	✓

Assumptions: A-11

Illustrative, not data:

- Faded network in the background is the slide 06 sample.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator (20,000 HCPs, 8 specialties, 12 regions, about 440 practice groups, 24 months of class prescriptions, 75,155 ties)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/hcp_targeting/data.py`; `scripts/generate_data.py`	make data; make run; export_call_plan.py + inject_sheet.py
A-02: The generator decides how much the network matters: conversion depends on hidden practice-group culture (sd 0	F-13, F-29, F-30, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-43, F-46, F-47, F-48, F-49, F-50	04, 06, 07, 08	`src/hcp_targeting/data.py:98, :188-189`	make run; export_call_plan.py + inject_sheet.py
A-03: A call budget is 5% (1,000 calls) or 20% (4,000 calls) of the 20,000 HCPs; 'converters reached' assumes the call is what converts the HCP (the model ranks propensity, not uplift from calling)	F-05, F-22, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-41, F-42, F-43, F-44, F-45, F-46, F-49	02, 05, 06, 07, 08	`src/hcp_targeting/cli.py:199-200`	make run; export_call_plan.py + inject_sheet.py
A-04: Label-derived features (community and neighbour conversion rates) are rebuilt inside every outer fold from training labels only, with an inner 5-fold split and a prior weight of 20; embeddings are trained without labels	F-10, F-11, F-12, F-13, F-27, F-35, F-40	03, 04, 05, 06, 07, 08	`src/hcp_targeting/features.py:153-184`; `src/hcp_targeting/evaluate.py:129 (prior_weight)`	make run; export_call_plan.py + inject_sheet.py
A-05: Louvain runs at resolution 1 (regional blocks) and 16 (practice-size communities); 16 was chosen to match practice size, not tuned on conversion	F-09, F-17, F-29, F-30, F-44	03, 05, 06, 07	`src/hcp_targeting/features.py:60`	make run; export_call_plan.py + inject_sheet.py
A-06: Betweenness is estimated from 256 sampled sources (64 in the seed sweep)	F-18, F-50	05, 08	`src/hcp_targeting/cli.py:198`	make run
A-07: Every arm uses one XGBoost setting (400 trees, depth 4, learning rate 0	F-13, F-19, F-20, F-32, F-33, F-34, F-35, F-36, F-37, F-38, F-39, F-46, F-47, F-48	04, 05, 06, 08	`src/hcp_targeting/evaluate.py:85-98`	make run; export_call_plan.py + inject_sheet.py
A-08: The shuffled-graph control moves every HCP to a random node of the same graph (seed 8 permutation): structure is unchanged, the person-position link is broken	F-11, F-38, F-48	03, 06, 08	`src/hcp_targeting/cli.py:93-104`	make run
A-09: The ceiling row scores HCPs by the generator's true propensity (answer key), which no feature reads	F-39, F-49	06, 08	`src/hcp_targeting/data.py`	make run
A-10: The call plan on slide 07 ranks the arm '+ community and neighbour conversion rates' (hand-built graph features, no embeddings) by its out-of-fold scores; the demo's export re-runs that arm and asserts AUC 0	F-40, F-41, F-42, F-43, F-44, F-45	07	`docs/demo/source/tools/export_call_plan.py (arm, CALLS)`	export_call_plan.py + inject_sheet.py; hyperframes render
A-11: Video rounding: AUC to 3 decimals, AUC deltas to 3 decimals with sign, percentages to 1 decimal (community rates, 8	F-05, F-08, F-30, F-46, F-49, F-50, F-51, F-52, F-53, F-54, F-55, F-56	02, 06, 08, 09	`the composition file named on each slide`	hyperframes render
A-12: Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, consistent with README 'The problem'	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

- All results come from one generated market (seed 7, plus two sweep seeds). The generator wires practice culture and opinion-leader exposure into conversion, so the size of the graph lift is set by the generator, not measured on a real market (README 'Known limitations', A-02).
- 'Converters reached' ranks propensity, not uplift from calling: there is no call response model (A-03).
- Slide 07 presents out-of-fold scores as a 'next cycle' plan. In the repo the snapshot is static: the same campaign window provides labels for the fold-built features and the outcomes being scored (README 'Known limitations').
- The slide 07 plan comes from the demo's export script, which re-runs one arm of make run (same data, seed, folds and settings) and asserts AUC 0.668 and 189 / 126 captured against outputs/ablation.csv. The repo itself does not write a plan file.

• 'Full run ≈ 403 s' quotes the README for an earlier session; the captured run in outputs/ took 12,279 s on a shared machine.

• Slide 09 shows the GraphSAGE delta as -0.002 and the both-embeddings delta as +0.005 (3-decimal rounding of outputs/deltas.csv); the README text writes -0.003 and +0.006.

• The 44% headroom figure uses the best arm (node2vec + GraphSAGE, 212); the +50% figure uses the hand-built graph-features arm (189).

• Slide 01 statements are general industry knowledge (A-12).

Appendix: environment

Item	Value
Commit	`719ffbfc4cf1d5bf57f19f9c80be25168ad972f6`
Python	3.11.7
xgboost	3.2.0
networkx	3.6.1
torch	2.10.0
torch_geometric	2.8.1
sklearn	1.9.1
pandas	3.0.6
numpy	2.4.6
Profile / seeds	data_seed=7, cv_folds=5, louvain_seed=7, xgboost_seed=7, bootstrap_resamples=200
Run time	summary.json runtime_seconds = 12,278.9 s (3.4 h) for the captured make run on a shared machine; README reports 403 s for a full run on an idle 10-core laptop. The demo's export re-run of one arm took about 30 s.